



# Stefano Magrini Alunno

## EDUCATION AND TRAINING

[ 1 Nov 2025 – Current ]

### PhD

**Università di Modena e Reggio-Emilia**

**City:** Ferrara | **Country:** Italy | **Field(s) of study:** Natural sciences, mathematics and statistics: • *Mathematics* | **Level in EQF:** EQF level 8

#### Skills Acquired

- Preparation of presentations and posters using LaTeX
- Numerical methods for kinetic equations

#### Conferences and Schools Attended / To Attend

- Particle Systems and PDEs XIII (December 1-5, 2025)
- DataHyKing Workshop (March 23-27, 2026)
- MFML Summer School (May 24-29, 2026)
- M&MKT Summer School (June 7-13, 2026)
- ICTP-INDAM-SLMATH Summer Graduate School on Machine Learning (June 15-26, 2026)
- LEAN Conference (September 7-11, 2026)
- HJE & MFG Workshop (October 1-2, 2026)

[ 1 Oct 2022 – 22 Jan 2025 ]

### Master's Degree

**Università degli Studi di Roma "La Sapienza"** <https://www.uniroma1.it/it/pagina-strutturale/home>

**City:** Rome | **Country:** Italy | **Field(s) of study:** Applied Mathematics | **Final grade:** 10/110 Cum laude | **Level in EQF:** EQF level 7 | **Number of credits:** 120 | **Thesis:** Exploring Authorial Identity through Data Mining Techniques

Curriculum aimed at Data Science with foundations in cryptography and computational mathematics.

Main courses followed:

Mathematical Statistics (lecturer: Gustavo Posta)

Stochastic Processes (lecturer: Lorenzo Taggi)

Combinatorics (lecturer: Claudia Malvenuto)

Mathematical Models for Neural Networks (lecturer: Elena Agliari)

Computational Mathematics (lecturer: Giuseppe Visconti)

Deep Learning and Applied Artificial Intelligence (lecturer: Emanuele Rodolà)

Other activities:

- University tutor

[ 24 Sep 2018 – 30 Apr 2022 ]

### Bachelor's Degree

**Università degli Studi di Roma "La Sapienza"** <https://www.uniroma1.it/it/pagina-strutturale/home>

**City:** Rome | **Country:** Italy | **Field(s) of study:** Natural sciences, mathematics and statistics: • *Mathematics* | **Final grade:** 110/110 | **Level in EQF:** EQF level 6 | **Number of credits:** 180 | **Thesis:** Analisi automatica di disegni per attribuzione d'autore

Curriculum aimed at numerical methods and elements of probability.

Main courses taken:

Geometry 1 (lecturer: Andrea Sambusetti)  
Algebra 1 (lecturer: Paolo Papi)  
Numerical Approximation Methods (lecturer: Giuseppe Visconti)  
Geometry 2 (lecturer: Andrea Sambusetti)  
Mathematical Analysis 1 (teacher: Fabiana Leoni)  
Probability 1 (lecturer: Gustavo Posta)  
Complex Variable (lecturer: Kieran Gregory O'Grady)  
Laboratory of Programming and Computation (lecturer: Maria Lopez Fernandez)  
Rational Mechanics (lecturer: Emanuele Caglioti)

## WORK EXPERIENCE

**Sapienza Aerospace Students Association** <https://www.sasa-aerospace.it>

**City:** Rome | **Country:** Italy | **Email address:** [sasa-administration@uniroma1.it](mailto:sasa-administration@uniroma1.it) | **Name of unit or department:** Computer Vision Division | **Business or sector:** Professional, scientific and technical activities

[ 15 Sep 2024 – 15 Aug 2025 ]

### Sapienza Flight Team

Role:

I was responsible for developing software capable of mapping the territory from photos taken at an altitude of 90m. Research and development activities were also carried out, trying to meet the hardware requirements imposed.

Activities performed:

Development of software capable of mapping a territory.

Creation of mathematical models (of machine learning, but also physical and optical)

Python programming

Report writing with LaTeX and Word.

Consulting in other departments that dealt with computer vision and autopilot software

Familiarity with Blender software, roboflow site and github.

**Sapienza Aerospace Student Association** <https://www.sasa-aerospace.it>

**City:** Rome | **Country:** Italy | **Email address:** [sasa-administration@uniroma1.it](mailto:sasa-administration@uniroma1.it) | **Name of unit or department:** AirDrop Division | **Business or sector:** Professional, scientific and technical activities

**Link** <https://www.youtube.com/watch?v=hb13KonX5lo&t=1s>

[ 15 Sep 2023 – 15 Aug 2024 ]

### Sapienza Flight Team

Role:

Initially a member of the computer vision department, then a member of the AirDrop department, and finally head of the AirDrop division.

Goals Achieved:

The department was responsible for designing a case (built with 3D printers) that could accommodate an 8 oz bottle, a parachute attachment of about 70cmx70cm, and a nano arduino connected to a 5V servo and powered by an 8.4V battery. The department, following numerous tests, was able to complete the work within the time allowed by the team leader by making an easy-to-use and safe prototype.

The team won the safety award of the SUAS 2024 international competition by placing 6th.

Activities performed:  
Organization and teamwork  
Creating mathematical models (of machine learning as well as physics and optics)  
C/C++, Fortran, Arduino and Python programming  
Organization of experiments plus collection and analysis of empirical data  
Report writing with LaTeX and Word  
Assistance in electronics department, computer vision and autopilot software  
Familiarity with Blender software, roboflow site and github.

**La Sapienza**    <https://www.uniroma1.it>

**City:** Rome   |   **Country:** Italy

[ 31 Aug 2024 – 30 Sep 2024 ]

### University Tutor

Preparation and exposure is in-person lectures. Use of blackboard, projector and slides for lecture delivery. Preparation, execution and correction of tests and exercises.

Topics covered: Algebra, Probability, Analysis and Geometry

Coordinator: prof. Andrea Sambusetti

## SKILLS

### Programming languages

Linux | R | Arduino | asmx86 | Python 3.10 | MATLAB 22b | ISO/IEC 14882:2011 (C++11) | Bash | ISO/IEC 9899:1999 (C99) | Fortran90 | Experience in GPU programming using CUDA C/C++.

### Computer skills

GoogleDrive | Microsoft(Microsoft Word, Microsoft PowerPoint, Microsoft Access, Microsoft Excel) | Github (Base)

## LANGUAGE SKILLS

**Mother tongue(s):** Italian

**Other language(s):**

**English**

**LISTENING** A2   **READING** B1   **WRITING** B1

**SPOKEN PRODUCTION** B1   **SPOKEN INTERACTION** B1

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

## CONFERENCES AND SEMINARS

[ 21 Jul 2019 – 24 Jul 2019 ]

### Summer School Perugia 2019 – INdAM Scholars Meeting

Department of Mathematics, University of Perugia

Summer school aimed at INdAM scholarship winners

[ 1 Dec 2025 – 5 Dec 2025 ]

### Conference PSPDE XIII    Modena

Auditor

**Link:** <https://sites.google.com/view/pspdexiii>

[ 23 Mar 2026 – 27 Mar 2026 ]

### Workshop DataHyking    Ferrara

Auditor

**Link:** [https://datahyking.eu/?page\\_id=928](https://datahyking.eu/?page_id=928)

[ 24 May 2026 – 29 May 2026 ]

### Summer School Mathematical Foundations of Machine Learning    Trento

Speaker: Lénaïc Chizat, Nicolás García-Trillos, Guido Montúfar, Enrique Zuazua

**Link:** <https://sites.google.com/unitn.it/mfml-2026>

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## HONOURS AND AWARDS

[ 31 May 2024 ]

**Team competition Awarding institution:** SUAS competition

It reached the 6th position in the world scale.

The competition required the development of a UAV that would accomplish a series of tasks within constraints given by the competition.

**Link:** [https://docs.google.com/spreadsheets/d/e/](https://docs.google.com/spreadsheets/d/e/2PACX-1vSV0_wulZ_djNrAeuGUOWQXyDoUyD6lgD5H7W9tqT8LZu_uv6s00iEsDP0lbRetUTWTqQ9S1RAdvAr/pubhtml?usp=embed_facebook)

[2PACX-1vSV0\\_wulZ\\_djNrAeuGUOWQXyDoUyD6lgD5H7W9tqT8LZu\\_uv6s00iEsDP0lbRetUTWTqQ9S1RAdvAr/pubhtml?usp=embed\\_facebook](https://docs.google.com/spreadsheets/d/e/2PACX-1vSV0_wulZ_djNrAeuGUOWQXyDoUyD6lgD5H7W9tqT8LZu_uv6s00iEsDP0lbRetUTWTqQ9S1RAdvAr/pubhtml?usp=embed_facebook)

[ 30 Sep 2018 ]

**INDAM Scholarship Awarding institution:** Istituto Nazionale Di Alta Matematica "Francesco Severi"

He reached the third national ranking following a test.

The scholarship was then renewed annually on merit until its natural end, which is after the third award.

**Link:** <https://www.altamatematica.it/wp-content/uploads/2018/05/Graduatoria402018-1.pdf>

[ 30 Apr 2018 ]

**Math Olympiad team competition Awarding institution:** Progetto Olimpiadi di Matematica

Achieved 10th national team ranking.

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**Link:** [http://olimpiadi.dm.unibo.it/wp-content/uploads/2021/06/GareASquadreDiMatematica\\_Cesenatico2018.pdf](http://olimpiadi.dm.unibo.it/wp-content/uploads/2021/06/GareASquadreDiMatematica_Cesenatico2018.pdf)

[ 30 Apr 2017 ]

**Bocconi individual national competition Awarding institution:** Università Bocconi - Giochi Matematici

He reached 44-th individual national ranking.

**Link:** <https://giochimatematici.unibocconi.it/index.php/archivio-giochi/finale-nazionale-2017-classifica-l1/viewdocument/93>

[ 30 Sep 2022 ]

**Member of MENSA association Awarding institution:** MENSA Italia

Recognition came as a result of an IQ test that you can only pass if you have passed the Mensa threshold of the 98-percentile of the population.Arseclamation

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## NETWORKS AND MEMBERSHIPS

[ 30 Sep 2022 – 30 Sep 2024 ]

**MENSA Italia** Italy

**Link:** <https://www.mensa.it/>

[ 1 Sep 2023 – 31 Aug 2025 ]

**Sapienza Aerospace Student Association (SASA)** Rome

Student Association of the Department of Aerospace Engineering at La Sapienza.

**Link:** <https://www.sasa-aerospace.it/>

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